THEIVANAI AMMAL COLLEGE FOR WOMEN (AUTONOMOUS)

(Affiliated to Annamalai University, Chidambaram - Tamil Nadu)
(Accredited by NAAC (4th Cycle) at 'A+' Grade)
(Recognized Under 2(f) and 12(B) by UGC)
Villupuram - 605 403
NOV-2026

PG & RESEARCH DEPARTMENT OF COMPUTER SCIENCE

Register No. | **20525PCSC004**

Certified that ***SARANYA M*** *of* ***M.Sc Computer Science, [II-Year]*** *has completed the* ***ADVANCED ARTIFICIAL INTELLIGENCE & MACHINE LEARNING PRACTICAL – PCSR309*** *by undergoing the practical as per the syllabus during the academic year 2026 – 2027.*

Course In-charge

MS. K. MANOHARI

Head of the Department of Computer Science

MS. K. MANOHARI

Date: 21.10.2026

CONTENTS

S.NO	DATE	TITLE
1	18.06.2026	IMPLEMENTATION OF UNINFORMED SEARCH ALGORITHMS (BFS, DFS) BFS: 1. Hospital Emergency Route Finding Using Breadth-First Search (BFS) Algorithm 2. Airport Passenger Boarding Order Using Breadth-First Search (BFS) Algorithm DFS: 1. Bakery Product Distribution Using Depth-First Search (DFS) Algorithm 2. Graph Traversal Using Depth-First Search (DFS) Algorithm
2	23.06.2026	IMPLEMENTATION OF INFORMED SEARCH ALGORITHMS (A*, MEMORY-BOUNDED A*) 1. Robot Battery Level Monitoring and Optimal Robot Selection 2. Treasure Hunting Using A* Search Algorithm
3	07.07.2026	IMPLEMENT NAÏVE BAYES MODELS 1. Email Spam Detection Using Naive Bayes Classification Algorithm 2. Loan Approval Prediction Using Naive Bayes Classification Algorithm
4	10.07.2026	IMPLEMENT BAYESIAN NETWORKS 1. Courier Package Loss Prediction Using Bayes' Theorem 2. Machine Failure Prediction Using Bayes' Theorem
5	13.08.2026	BUILD REGRESSION MODELS 1. Electricity Bill Calculation and Visualization 2. Battery Backup Time Prediction Using Linear Regression Algorithm
6	13.08.2026	BUILD DECISION TREES AND RANDOM FORESTS DECISION TREE: 1. Weather Prediction Using Decision Tree Classification and Data Visualization 2. Email Spam Detection Using Decision Tree Classification Algorithm

S.NO	DATE	TITLE
		RANDOM FORESTS: 1. Traffic Accident Severity Prediction Using Random Forest Classification Algorithm 2. Sentiment Analysis of Product Reviews Using Random Forest Classification Algorithm
7	07.09.2026	BUILD SVM MODELS 1. Fruit Classification Using Support Vector Machine (SVM) Algorithm 2. Electricity Usage Prediction Using Support Vector Regression (SVM) Algorithm
8	24.09.2026	IMPLEMENT ENSEMBLING ALGORITHMS 1. Student Mark Prediction Using Ensemble Learning (Voting and Averaging) Algorithm 2. Water Pipeline Leakage Risk Prediction Using Ensemble Learning (Voting Classifier) Algorithm
9	28.09.2026	IMPLEMENT CLUSTERING ALGORITHMS 1. Bank Transaction Amount Classification Using K-Means Clustering Algorithm 2. User Behavior Grouping Using K-Means Clustering Algorithm
10	28.09.2026	IMPLEMENT EM FOR BAYESIAN NETWORKS 1. Traffic Jam Prediction Using Expectation-Maximization (EM) Algorithm 2. Machine Failure Risk Prediction Using Expectation-Maximization (EM) Algorithm
11	05.10.2026	BUILD SIMPLE NN MODELS 1. Delivery Time Prediction Using Sigmoid Activation Function 2. Traffic Congestion Classification Using Neural Network (NN)
12	05.10.2026	BUILD DEEP LEARNING NN MODELS 1. Loan Approval Prediction Using Deep Learning Neural Network 2. Equipment Failure Prediction Using Sigmoid-Based Neural Network Classification